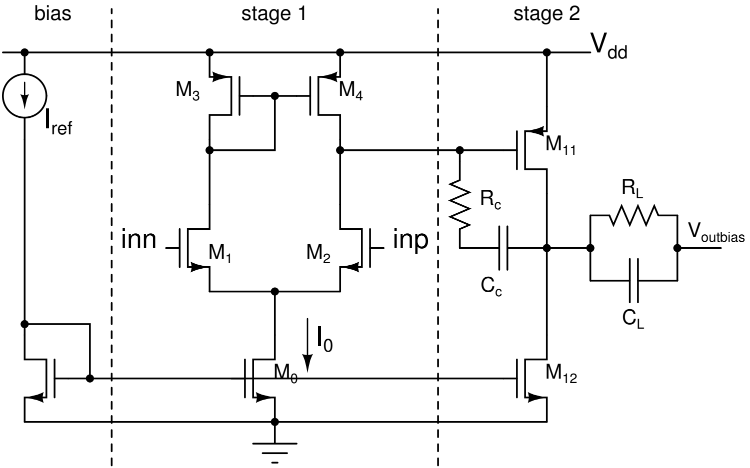
Two stage opamp



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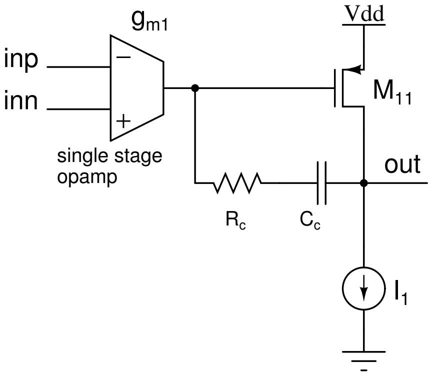

- First stage can be Differential pair, Telescopic cascode, or Folded cascode; Ideal g_{m1} assumed in the analysis
- Second stage: Common source amplifier
- Frequency response is the product of frequency responses of the first stage g_m and a common source amplifier driven from a current source

Common source amplifier: Frequency response

$$
\begin{aligned}
\frac{V_o(s)}{V_d(s)} &= \left(\frac{g_{m1} g_{m11}}{G_1 G_L}\right) \frac{sC_c(R_c - 1/g_{m11}) + 1}{a_3 s^3 + a_2 s^2 + a_1 s + 1} \\
a_3 &= \frac{R_c C_1 C_L C_c}{G_1 G_L} \\
a_2 &= \frac{C_1 C_c + C_c C_L + C_L C_1 + R_c C_c (G_1 C_L + C_1 G_L)}{G_1 G_L} \\
a_1 &= \frac{C_c (g_{m11} + G_1 + G_L + G_1 G_L R_c) + C_1 G_L + G_1 C_L}{G_1 G_L}
\end{aligned}
$$

- G_1: Total conductive load at the input
- G_L: Total conductive load at the output
- C_1: Total capacitive load at the input
- C_L: Total capacitive load at the output

Common source amplifier: Poles and zeros

$$
\begin{aligned}
p_1 &\approx -\frac{G_1}{C_c(\frac{g_{m11}}{G_L}+1+\frac{G_1}{G_L}+G_1R_c)+C_1(1+\frac{G_1}{G_L})} \\
p_2 &\approx -\frac{g_{m11}\frac{C_c}{C_1+C_c}+G_L+G_1\frac{C_c+C_L}{C_1+C_C}+G1G_LR_c\frac{C_c}{C_1+C_c}}{\frac{C_1C_c}{C_1+C_c}+C_L+\frac{R_cC_c(G_1C_L+G_LC_1)}{C_c+C_L}} \\
p_3 &\approx -\left(\frac{1}{R_c}\left(\frac{1}{C_L}+\frac{1}{C_c}+\frac{1}{C_1}\right)+\frac{G_1}{C_1}+\frac{G_L}{C_L}\right) \\
z_1 &= \frac{1}{(1/g_{m11}-R_c)C_c}
\end{aligned}
$$

Unity gain frequency

$$
\omega_u \approx \frac{g_{m1}}{C_c\left(1+\frac{G_L}{g_{m11}}+\frac{G_1}{g_{m11}}+\frac{G_1G_LR_c}{g_{m11}}\right)+C_1\left(\frac{G_L}{g_{m11}}+\frac{G_1}{g_{m11}}\right)}
$$

- Pole splitting using compensation capacitor C_c
  - p_1 moves to a lower frequency
  - p_2 moves to a higher frequency (For large C_c, $p_2 = g_{m11}/C_L$)
- Zero cancelling resistor R_c moves z_1 towards the left half s plane and results in a third pole p_3
  - z_1 can be moved to ∞ with $R_c = 1/g_{m11}$
  - z_1 can be moved to cancel p_2 with $R_c > 1/g_{m11}$ (needs to be verified against process variations)
  - Third pole p_3 at a high frequency
- Poles and zeros from the first stage will appear in the frequency response—$Y_{m1}(s)$ instead of g_{m1} in V_o/V_i above
  - Mirror pole and zero
  - Poles due to cascode amplifiers

Compensation cap sizing

$$p_2 \approx -\frac{g_{m11}\frac{C_c}{C_1+C_c}}{\frac{C_1 C_c}{C_1+C_C} + C_L}$$

$$\omega_u \approx \frac{g_{m1}}{C_c}$$

Phase margin (Ignoring $p_3, z_1, \ldots$)

$$\phi_M = \tan^{-1}\frac{|p_2|}{\omega_u}$$

$$\frac{|p_2|}{\omega_u} = \tan\phi_M$$

$$\frac{g_{m11}}{g_{m1}}\left(\frac{C_c}{C_L}\right)^2 = \frac{C_c}{C_L}\left(1+\frac{C_1}{C_L}\right)\tan\phi_M + \frac{C_1}{C_L}\tan\phi_M$$

- For a given ϕ_M, solve the quadratic to obtain C_c/C_L
- If C_1 is very small, $p_2 \approx -g_{m2}/C_L$; further simplifies calculations

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A_o	$g_{m1}g_{m11}/(g_{ds1}+g_{ds3})(g_{ds11}+g_{ds12})$
A_{cm}	$g_{ds0}g_{m11}/2g_{m3}(g_{ds11}+g_{ds12})$
C_i	$C_{gs1}/2$
ω_u	g_{m1}/C_c
p_k, z_k	See previous pages
S_{vi}	$\approx 16kT/3g_{m1}\,(1+g_{m3}/g_{m1})$
σ^2_{Vos}	$\approx \sigma^2_{VT1}+(g_{m3}/g_{m1})^2\sigma^2_{VT3}$
V_{cm}	$\geq V_{T1}+V_{DSAT1}+V_{DSAT0}$ $\leq V_{dd}-V_{DSAT3}-V_{T3}+V_{T1}$
V_{out}	$\geq V_{DSAT12}$ $\leq V_{dd}-V_{DSAT11}$
SR+	I_0/C_c
SR-	$\min\{I_0/C_c, I_1/(C_L+C_c)\}$
I_{supply}	$I_0+I_1+I_{ref}$